



Prepared with Synoptic

Final investigation report

Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads

Date of issue

08/10/2026

Contents

Executive summary	3
What happened.....	3
What the investigation found.....	3
What can be done.....	3
Factual information	5
History of the occurrence.....	5
Personnel information	6
Vessel and equipment information.....	6
Meteorological information	6
Recorded data.....	6
Channel and site information	7
Organisational and management information	7
Tests and research.....	8
Analysis	9
External Influences	9
Organisational Influences.....	10
Risk Controls.....	11
Local Conditions	13
Individual Actions	15
Occurrence Events.....	17
Conclusions	19
Findings	19
Probable cause	20
Contributing factors	20
Other factors that increased risk	21
Other findings.....	21
Recommendations.....	23
Clearing of the beacon's sightline deferred unfunded for three works rounds	23
Last service moved into darkness without reopening the daylight-only risk assessment	23
No night passage plan or hold point for loss of the primary fix	23
Defect procedure never requires a fault that cleared itself to be recorded.....	24
Aids code set no notice or availability duty for beacons off declared channels	24
Beacon's range halved by solar conversion with no notice to channel users	24
Darkness transit rostered to close a split duty within 20 minutes of the spread limit	25
Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey	25
Distribution	26

Executive summary

What happened

On 07/07/2026 the 24.4 m aluminium catamaran passenger ferry MV Tarrakin Star was inbound through the Merrandi Channel in darkness on service TF18. It carried 38 passengers, a master, a deckhand and a kiosk attendant. At 18:07:06 the chart plotter display froze and did not recover. The heading eased from 195 to 200 degrees and the vessel left the dredged channel to the west. At 18:12:04 it made contact with Boolarong Bank at 6.8 kt, 74 m west of the charted centreline and 54 m outside the dredged channel. It stopped within 11 m on a falling tide.

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The starboard hull bottom plating was set up over 4.1 m with a 180 mm split, and the No. 2 void flooded to approximately 260 mm. The starboard propeller, shaft, rudder stock and skeg were damaged. All passengers were ashore by 19:26. The vessel refloated at 00:20 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

What the investigation found

The probable cause was the continuation of the transit at about eight knots after the chart plotter froze, with no night passage plan or hold point for loss of the primary fix. This allowed a five degree heading change, made without a decision to turn, to take the vessel onto the bank.

The freeze very likely contributed. Firmware issued in 11/2024 addressed the fault but had never been applied. The defect procedure did not require self-cleared faults to be recorded, so none of the freezes masters described over the previous year was written down. This likely contributed, although how quickly a recorded freeze would have led to the fix was not established.

Nothing on the bridge showed the master the heading change. Mangroves on Merrandi Islet screened the No. 4 beacon, the one reference left, between 0.84 nm and 0.34 nm north of it. This likely contributed by letting the turn go unnoticed. The Port Authority's deferral of clearing that sightline as unfunded at three works rounds very likely contributed.

The vessel continued at speed until the shallow alarm sounded, although it could have been stopped in 92 m on flat water with no other traffic. This very likely contributed. The absence of a hold point very likely contributed. Moving TF18 into darkness without reopening the daylight-only risk assessment likely contributed by leaving no night passage plan. Why the change was assessed as a timetable change only was not established. The shallow alarm, still at its 2019 setting of 0.9 m, gave 16 seconds' warning against a 92 m crash stop, which very likely contributed.

A tidal set-down and a steering or propulsion failure were both very unlikely. Whether floodlights behind the beacon delayed the sighting, and whether the master was fatigued, could not be determined. After contact the master stopped the engines and decided within a minute not to refloat the holed vessel, which very likely reduced risk.

What can be done

The investigation recommended that Tarrakin Ferries write a night passage plan for the channel before any further darkness transit. The plan should name a hold point requiring way to be taken off when the primary fix is lost inside the channel. No action had been recorded against the risk assessment process or the defect procedure.

The Merrandi Islet canopy was trimmed to 2 m on 24/07/2026 and a three-yearly cycle was programmed. The investigation recommended that the services agreement replace its 50 m vegetation scope with a sightline test.

Four further safety issues increased risk without contributing. The aids code gave users of an undeclared channel no right to notice of changes, which likely increased risk. The beacon's solar conversion halved its nominal range without notice, which likely increased risk. In response, LNM-2026-14 was issued, the aids register was opened to operators and the passage plan appendix was corrected. The winter roster placed the darkness transit 20 minutes inside the spread limit, which likely increased risk. A kiosk counter, also fitted to two other vessels, blocked the forward lifejacket locker, which very likely increased risk. No action had been recorded against the code, the roster or the locker.

Factual information

History of the occurrence

On 07/07/2026 MV Tarrakin Star was operating TF18, the last service of the day from Muntari Island to the Kanooka Heads town wharf. At about 13:05 the relieving master took over from the morning master at the town wharf. The outgoing master said the chart plotter had frozen twice that morning and recovered by itself both times. The relieving master said it had done the same to him the month before. Neither entered the fault in the wheelhouse defect book.

TF18 departed at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant. Sunset was at 17:02 and civil twilight ended at 17:29. The occurrence notification states that "the channel transit was made in full darkness." The winter timetable had placed this service in darkness since 01/06/2026, and this was the thirty-first such transit.

The vessel entered the Merrandi Channel from the north at 18:03:58. At 18:04:14 the master sent the deckhand to collect fares and was then alone on the bridge. At 18:05:30 the vessel was on the charted centreline abeam the No. 2 beacon, making 8.4 kt on 195 degrees.

At 18:07:06 the chart plotter display stopped responding. The chart stayed on screen, but the vessel symbol stopped moving and touch input was not accepted. At 18:07:11 the master said "Oh, come on," and sounds consistent with repeated screen taps followed. At 18:07:44 he said "(unintelligible) frozen again." Unlike that morning, the unit did not recover. In the same second the heading began to increase from 195 degrees, reaching 199 degrees at 18:08:31, where it was held.

At 18:08:31 the master said "Right. Where's four." He was looking for the No. 4 starboard-hand beacon on the western side of the channel. The vessel was then 12 m west of the centreline and 0.47 nm north of the beacon. It crossed the western edge of the dredged channel at about 18:09. At 18:09:40 a sound consistent with a window or door sliding open was recorded. At 18:10:02 the master said "There's four. Bring her back," with the vessel 39 m west of the centreline and 0.25 nm north of the beacon. The course over the ground eased from 200 to 197 degrees but stayed west of the channel's 195 degrees until contact.

The echo sounder shallow alarm sounded at 18:11:48 with 0.9 m under the keel. Both throttles moved to astern together four seconds later. At 18:12:04 the vessel made contact with Boolarong Bank at 6.8 kt over the ground, 74 m west of the charted centreline and 54 m outside the dredged channel. It stopped within 11 m. The notification records it as "heeled 4 degrees to starboard and hard aground on a falling tide." The master's broadcast described the vessel as "stable" and "upright".

Both engines were stopped at 18:12:31 and not restarted. At 18:13:04 the master broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16. He gave the position, persons on board, no injuries and no pollution, and stated "I am not attempting to refloat." Port Control called out the port workboat. At 18:14:52 the master told passengers that lifejackets were being issued. At 18:16 the deckhand sounded the No. 2 starboard void and found about 75 mm of water, then sounded it every ten minutes. All passengers were in lifejackets by about 18:22. In the forward saloon the deckhand had to pass behind the kiosk counter to reach the locker. The workboat came alongside at 18:41 and the last passenger was ashore at 19:26.

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The vessel refloated on the rising tide at 00:20 on 08/07/2026 with approximately 260 mm of water in the No. 2 void. It was towed to the town wharf and withdrawn from service. The Merrandi Channel was closed to commercial traffic for 19 hours.

Personnel information

The master had about 19 years in commercial marine operations, 14 as a master and 11 on this route, with about 4,100 Merrandi Channel transits. He had held a master's certificate for sheltered-waters passenger vessels since 2012. His last route check, on 14/02/2026, was satisfactory. He had made 26 of the earlier darkness transits.

He was on day nine of a twelve-day tour, working a split duty from 05:40 to 08:40 on another of the operator's vessels and from 13:00 to 19:20. He had slept 6 h 15 min the night before. He said he was tired but not fighting to stay awake and not slow, which he described as normal for a split. [NOT ESTABLISHED: the operator's split duty spread limit and the margin to it on 07/07/2026].

The outgoing master had 9 years as a master, 4 on this route, and about 1,400 channel transits. His last route check, on 09/11/2025, was satisfactory. He worked 05:40 to 13:05 on day six of a twelve-day tour and had made 4 of the darkness transits.

The deckhand had 7 months with the operator and was in a first winter season on the route. The deckhand held a basic marine safety certificate and a coxswain's certificate and had completed vessel familiarisation and drills in 12/2025, but had never been shown the channel marks.

Vessel and equipment information

MV Tarrakin Star is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers. It completed its annual survey on 11/03/2026 with no outstanding conditions and was last slipped in 03/2025. None of the eleven defect book entries in the preceding 90 days concerned the chart plotter, GPS antenna, echo sounder, steering gear, engines or propulsion controls. The maintenance system held no work order, fault report or correspondence on the plotter in 2026.

The chart plotter is a 12-inch multifunction display fed by a separate GPS antenna unit. It ran chart software revision 4.11 on firmware issued 03/2024. Firmware issued in 11/2024 addresses cause code 0x51 and had not been applied. The operator did not hold the manufacturer's bulletin and had no subscription to its bulletins. The outgoing master believed the plotter had frozen three or four times in the past year, none written up.

The shallow alarm had been set at 0.9 m under the keel at commissioning in 2019 and not changed. The vessel's trials figure for a crash stop from 8 kt is 92 m.

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon locker holds 40 adult jackets and is mounted behind the kiosk servery counter. The counter was fitted in a 2019 refit under a minor modification notice, with no re-survey of the safety equipment arrangement. The 2011 build arrangement had an open bench there with the locker face clear. The same counter is fitted to the operator's two other 24 m vessels.

The starboard hull bottom plating was set up over 4.1 m between frames 6 and 11, with a 180 mm split in way of the No. 2 void. All three starboard propeller blades were bent aft and the shaft was bent 1.4 mm. The starboard rudder stock was distorted 3 degrees and the starboard skeg was torn away. The port hull, propulsion and steering were undamaged.

Meteorological information

The master described the conditions as clear, with a light westerly and no sea in the entrance.

Recorded data

Times are local (UTC plus 10:00). The plotter and echo sounder logs were corrected by 2 seconds to the AIS receiver clock.

The plotter event log recorded watchdog resets with cause code 0x51 at 10:42 and 11:17. At 18:07:06 it recorded the display process not responding, with no reset. Power was removed at 18:16.

The AIS track showed the rate of turn peaking at 6 degrees per minute at 18:07:40 and returning to zero by 18:08:50. On the new heading the vessel made about 20 m of westing a minute. Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement.

The engines ran at 1,780 and 1,784 rpm until 18:11:52 with no alarm. Steering hydraulic pressure stayed at 96 to 99 bar against a normal band of 90 to 105 bar, and rudder angle followed helm demand within 1.5 degrees. At contact, astern thrust was developing but had not reached full. The shallow alarm sounded once, and the vessel covered 58 m in the 16 seconds before contact.

Channel and site information

The dredged channel has a charted half width of 20 m. The grounding position was 34 m west of the No. 4 beacon. The bed at the contact point was 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west, a gradient of about 1 in 70. Furrows ran 11 m from first contact on a heading of 197 degrees.

The No. 4 beacon shows Fl G 2.5s from a lantern 4.2 m above chart datum. In 11/2025 it was converted from mains to a solar lantern, and its nominal range changed from 6 nm to 3 nm. The port notified the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no channel user was told. The operator's passage plan appendix still gave the range as 6 M.

Merrandi Islet, about 10 m across and carrying mangroves, lies about 240 m north of the beacon in the shoal strip west of the dredged edge. Its canopy was measured at 5.8 m, against 2.1 m in the Council's 2019 aerial survey.

Both masters said shore lighting behind the beacon had made it harder to see. The master said twice that he could not say whether the lights were why he could not see it. [NOT ESTABLISHED: position, orientation and installation date of the shore floodlights.]

Organisational and management information

The operator is Tarrakin Ferries. The route's risk assessment covered daylight operation and was not reopened when TF18 moved into darkness [NOT ESTABLISHED: date and content of the risk assessment]. The master said the passage plan stated that night transits were not scheduled. There was no night passage plan and no hold point for loss of the primary fix. The passage plan named a speed, a course and three beacons. Neither it nor OPS-3 addressed who, or a second person, was on the bridge. Both masters and the deckhand said the timetable change came with a new roster and nothing else.

The defect procedure did not require a self-cleared fault to be recorded. The master described the practice as writing up what was broken and passing on verbally what had come good.

The operator's reporting system had a navigation hazard category, unused on this route in three years, and no master interviewed knew it existed. Both masters said the beacon had become harder to see over two years and neither had reported it. The master said "There's no form for the beacon looks dim."

The Port Authority raised work order WO-118 in 09/2023 to trim the islet's mangroves, estimated at \$18,500. It was deferred as unfunded at the 2023, 2024 and 2025 works rounds. WO-176, raised in 03/2026, recorded the sightline as fully closed on the northern approach. The Authority's file note placed the islet outside the 50 m scope of schedule 2.4 of the services agreement, so the work

fell to the general waterways vote. The monthly check recorded the beacon as observed and operating from 01/2024 to 07/2026. It was made from the town wharf, 2.1 nm south, where the islet is not in the line of sight. No defect or notice was ever raised against the beacon, and the aids register was never provided to Tarrakin Ferries.

Tests and research

The steering gear was removed and tested on 12/07/2026. All of it was serviceable and it passed a lock-to-lock timing test. The plotter was bench tested over 46 hours from 21/07/2026 and the fault did not recur.

A range and visibility check from the inbound track on 18/07/2026 measured the lantern's luminous range at 2.6 nm. The light was not visible from the channel between 0.84 nm and 0.34 nm north of the beacon. A vessel west of the track would see it open earlier. On 07/07/2026 the vessel passed about 24 m west of the islet at about 18:11.

A stowage survey on 16/07/2026 found the forward locker could not be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap. The muster placard directs passengers to that locker, which is not visible from the seats.

Analysis

External Influences

Aids code set no notice or availability duty for beacons off declared channels

The Waterways Aids to Navigation Code WAN-2 sorts aids by the status of the waterway on which they stand. Section 4.2 applies only "Where an aid to navigation is established on a declared shipping channel". For such an aid, the authority "must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect."

The Merrandi Channel is not a declared shipping channel, so section 4.3 applied to all six of its beacons instead. That section requires only that the authority "must be able to demonstrate that the aid is inspected at intervals it has determined and recorded." The services agreement sets that interval, but it adds nothing on notice. Indeed, schedule 2 states that "There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." It is virtually certain that this gap existed, because the code's text and the channel's status are documentary and were not in dispute.

The case for influence rests on two mechanisms. Had section 4.2 applied, the solar conversion of 11/2025 would have gone out as a notice to mariners before it took effect, and the Authority would also have been recording the No. 4 beacon's availability. These are the only means in the code by which anything about the beacon would have reached the people using the channel. Neither existed for it. No local notice was issued, no operator was told, and on the night the operator's passage plan appendix still showed the beacon at 6 M.

Neither mechanism, however, reaches this occurrence. A notice of the range change would have described a light that could not be seen from the channel between 0.84 nm and 0.34 nm north of it, and that was the stretch over which the vessel left the channel. Knowing a light's range better does not make it visible through a mangrove canopy. The same point decides the solar conversion itself, which was found not to have influenced the grounding and is dealt with under Organisational Influences.

An availability figure, for its part, measures whether a light is lit. The No. 4 beacon was lit throughout, and the monthly log showed it as operating in every month from 01/2024 to 07/2026.

Nor was the code the reason the screening remained. Section 4.1 has no channel threshold, so it applied to this beacon whatever the channel's status. The Authority's own register had recorded the screening since 09/2023 under WO-118. The work stayed unfunded because of the 50 m vegetation scope at schedule 2.4 of the services agreement. Merrandi Islet stands 240 m from the beacon, which placed the work in the general waterways vote. That deferral is argued under Organisational Influences, and the screening under Local Conditions.

It is therefore unlikely that the absence of notice and availability duties influenced the grounding. The evidence does not leave the question open, because each mechanism the code withholds meets a fact that settles against it.

The gap still matters for the channel, which carries a 96-passenger ferry in darkness. Under section 4.3 the operator has no right to be told when a beacon's range, character or position changes, and has no figure for how often the aids work.

The solar conversion shows the effect. It halved the No. 4 beacon's nominal range from 6 to 3 nautical miles, yet in the eight months before the occurrence it reached nobody who used the channel. A notification was lodged with the hydrographic service on 04/02/2026, but the chart

correction was still pending on 07/07/2026. The aids to navigation register has never been provided to the operator.

Nor does the monthly confirmation add anything for users. It is made from the town wharf, 2.1 nm south of the beacon, and nothing in the agreement requires any aid to be observed from the approach a vessel actually uses.

A future change to a beacon that is not screened would therefore reach the channel's users no more readily than this one did. The Authority can close the gap by its own practice, whatever the code requires.

The absence of any notice or availability duty for beacons off declared channels did not influence this occurrence, but it left the channel's users without information about its aids and was likely to be important to safety on the channel.

Organisational Influences

Last service moved into darkness without reopening the daylight-only risk assessment

CHG-2026-014 asked whether the winter timetable introduced anything new. It did not ask whether RRA-07 still held. The change was assessed as a "timetable change only", recorded "No new hazard identified" and answered "No" to route risk assessment review. The Operations Manager approved it, although he sits one level below the General Manager Operations, whose approval RRA-07 requires for any night transit. The daylight restriction is not referenced anywhere in the change. As a result, RRA-07 was not reopened and no night passage plan was written. TF18 then crossed the channel in darkness 31 times, and the occurrence transit was the thirty-first.

The restriction states that a night transit "requires the General Manager Operations' approval and a specific passage plan." A review that read it could not have closed without such a plan. Moreover, RRA-07 already named the "single unlit reach between the No. 2 and No. 4 beacons" as a hazard, so a plan written for this channel at night is the document most likely to say what to do when the primary fix is lost. The absence of that plan and hold point is argued under Risk Controls.

Two matters weigh against this. First, a reopened assessment might have produced a night plan no fuller than the daylight one, with no hold point, and the vessel would then have met the frozen plotter in the same way. That goes to how good the plan would have been, however, not to whether there would have been one. Second, the Operations Manager gave his understanding of what the restriction required of him only verbally, so why the change was assessed as a timetable change was not established. That bears on how the decision was reached, not on whether it was made.

It was likely that approving the move of TF18 into darkness without reopening RRA-07 left the channel without a night passage plan.

Clearing of the beacon's sightline deferred unfunded for three works rounds

The Authority's register identified the problem in 09/2023. WO-118 recorded mangrove regrowth on Merrandi Islet screening the No. 4 beacon from the northern approach, recommended trimming to 2 m on a three-yearly cycle, and estimated \$18,500. It was deferred at the 2023, 2024 and 2025 works rounds, each time as unfunded and referred to the next round. By 03/2026, WO-176 recorded the sightline as fully closed on the northern approach.

The islet is 240 m from the beacon, outside the 50 m scope of schedule 2.4. The work therefore competed with boat ramps and pontoons in the general waterways vote, rather than with aids to navigation. Meanwhile the monthly check could not show the problem. It was made from the town

wharf, 2.1 nm south, where the islet is not in the line of sight, and it recorded the beacon as observed and operating every month.

The step from the deferral to the screening rests on the Authority's own figures. The canopy stood at 2.1 m in the 2019 survey and at 5.8 m on 18/07/2026, a growth of about half a metre a year. A trim funded at the 2023 round would have left it near 3.5 m on the night, about 0.7 m below the 4.2 m lantern rather than 1.6 m above it. The qualification is that growth may not have been even, since only two measurements and the work order's description exist across the seven years.

It was very likely that the deferral of WO-118 left the canopy above the lantern on the night, screening the beacon from the channel between 0.84 nm and 0.34 nm north of it.

Beacon's range halved by solar conversion with no notice to channel users

The No. 4 beacon was converted to a solar lantern in 11/2025, and its nominal range fell from 6 to 3 nautical miles. A notification reached the hydrographic service on 04/02/2026, but the chart correction was still pending on 07/07/2026. No local notice was issued and no operator was told, so the operator's passage plan still gave the light at 6 M. The luminous range was measured at 2.6 nm.

The master described the light as weak and recalled it as a former mains light. For at least the last 37 seconds of his search it was clear of the islet and not yet seen, and a weaker light is harder to pick up against a bright background. Against this, the beacon was screened over the half mile in which the vessel left the channel, and brightness does not make a screened light visible. The master also already knew the light was weaker, so a notice would have told him nothing new. His search ran from 0.47 nm to 0.25 nm, well inside the measured range, and nothing shows the 6 M figure was used on the night. The only remaining mechanism is the light's conspicuity against the boat ramp floodlights. That question is unresolved and is discussed under Local Conditions.

It was unlikely that the reduced range influenced the occurrence. The safety issue is that a change to an aid on a passenger route reached no user of the channel, and the operator's plan described a light that was no longer there.

Darkness transit rostered to close a split duty within 20 minutes of the spread limit

In the winter timetable, TF18 closes a split duty of 9 hours 20 minutes across a 13 hour 40 minute spread, against the operator's limits of 10 hours and 14 hours. The darkness transit therefore falls in the last hour and a quarter of the spread, on every day of a twelve-day tour. No relief master is rostered. The operator asks for a fatigue diary but does not collect it.

The roster could reach this occurrence only through fatigue. The master was on day nine, had recorded 6 hours 15 minutes of sleep, and described himself as tired. His recorded performance, however, shows no sign of fatigue. He held the centreline to within 3 m for the first two minutes in the channel and responded to the shallow alarm in 4 seconds. His radio work was complete and correct at the first attempt, and the decision not to refloat was prompt and sound. Both masters said that no document asked them to stop when the plotter froze. The outgoing master, coming off a morning shift, gave the same account of the handover and the beacon. Whether the master was fatigued is an unresolved matter, discussed under Local Conditions.

It was unlikely that the roster influenced the occurrence. The safety issue is that it places every winter darkness transit at the end of a near-limit spread, with no relief.

Risk Controls

No night passage plan or hold point for loss of the primary fix

The Merrandi Channel passage plan named the chart plotter as the primary means of position fixing, and the visual transit of the No. 4 beacon with the town wharf leading lights as the secondary means. It said nothing about what the master should do if the primary means was lost, and it named no abort criterion or hold point anywhere. The plan closed by stating that it was written for the daylight transit and that night transits were not scheduled. No night plan existed to take its place.

Where it was needed, the secondary means offered little. The leading lights lie about 1.9 nm beyond the No. 6 beacon and do not resolve as two lights until a vessel is inside the channel's southern third. So when the plotter froze, 0.68 nm north of the No. 4 beacon, the beacon was the whole of the secondary means. Its screening from the channel is analysed under Local Conditions. OPS-3 3.6 requires a reduction of speed and consideration of anchoring, but only in reduced visibility. Visibility was not reduced, so that provision did not apply.

There was time to stop. The master had 114 seconds between the freeze at 18:07:06 and the vessel crossing the channel's western edge at about 18:09. A crash stop begun before about 18:11:40 would still have stopped the vessel short of the contact point. The water was flat, there was no other traffic, and a named hold point would have applied throughout. Both masters interviewed said independently that the right response was to stop, and that nothing told them to. Even so, a master who did not think of stopping unprompted might not have applied a written rule under the same pressure, and that is why the influence is put no higher than very likely. The decision to continue is analysed under Individual Actions, and the change that put the transit into darkness under Organisational Influences.

The absence of a night passage plan and of any hold point for loss of the primary means of position fixing very likely contributed to the transit continuing at eight knots after the plotter froze.

Defect procedure never requires a fault that cleared itself to be recorded

OPS-9 gives an intermittent fault on a non-statutory item no point at which it must be written down. 9.1 requires immediate reporting only of a defect that renders the vessel unseaworthy or affects a statutory item. The chart plotter is not a statutory item on this vessel on this route, even though the passage plan names it as the primary means of position fixing. 9.2 requires other defects to be entered at the end of the shift in which they are found, so a fault that has come and gone by then meets no trigger. The manual does not say whether such a fault is a defect. Finally, 9.3 requires a change-of-master brief but not a written one, and requires nothing of the relieving master afterwards.

Both masters described the same reading independently. A fault still present at handover goes in the book, while one that has cleared is passed on by word of mouth. The defect book bears this out, since its eleven entries in the 90 days before the occurrence were all faults present when they were written.

On the outgoing master's account, the plotter froze three or four times on this vessel in the year before the occurrence. It froze on the relieving master about a month earlier, and it froze twice on the morning of 07/07/2026, at 10:42 and 11:17. None of those freezes was written down. Had they been recorded, they would have reached the Operations Manager's weekly review and the maintenance system. The contractor identified cause code 0x51 against the 11/2024 firmware in a single bench test, and that firmware had not been applied.

Against this, the path from a written entry to an applied fix ran through somebody ashore finding the manufacturer's bulletin, and the operator held no subscription to the bulletins. A recorded freeze might therefore have been reviewed, noted and left. How many recorded freezes it would have taken for the unit to reach the contractor, and how quickly, could not be established. Even so,

a year of recorded freezes on the primary means of position fixing was unlikely to have gone to nobody.

The verbal handover at 13:05 without a defect book entry was one instance of this procedure and is analysed under Individual Actions. The procedure explains that omission, but it does not excuse it.

The absence of any requirement to record a fault that had cleared itself likely contributed to the plotter display freezing during the transit, although how quickly a recorded fault would have led to a fix could not be established.

Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey

The forward saloon's locker holds 40 adult lifejackets in the bulkhead behind the kiosk serving counter. Its face cannot be reached from 22 of the saloon's 34 seats without passing behind the counter through a 0.6 m gap at its inboard end. With people standing at the counter, as there were on the night, that gap is the only route. The muster placard directs passengers to a locker they cannot see from their seats. The arrangement approved at build in 2011 had an open bench in that position. The counter was fitted in a 2019 refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement.

The locker played no part in the grounding or its outcome. All 38 passengers were in lifejackets within about eight minutes and ashore within 74 minutes. The vessel was aground, upright and stable, and the muster took a couple of minutes longer than it should have. In other circumstances, though, a locker that passengers cannot reach without crew help would matter far more: in a fire, with a capsize risk, in a night evacuation into the water, or with a crew member injured in the impact. The locker remains behind the counter on this vessel, and the operator advised that the same counter is fitted to its two other 24 m vessels.

The obstruction of the forward saloon lifejacket locker very likely increased risk.

Local Conditions

Chart plotter display froze during the channel transit and did not recover

Three independent sources agree to the second that the display froze.

- The plotter's event log showed the display process stopped responding at 18:07:06, with no reset.
- The bridge audio has "Oh, come on" at 18:07:11 and "(unintelligible) frozen again" at 18:07:44.
- The master's account was that the chart stayed on the screen, the vessel symbol was stationary and the display would not accept touch input.

The display was virtually certain to have frozen.

The 46 hour bench test did not reproduce the fault and found no defect, but that result bears on the cause of the freeze rather than on whether it occurred. The cause code was 0x51, the same code logged at 10:42 and 11:17 that morning. The contractor identified it as a known fault addressed by firmware issued in 11/2024, which had never been applied to the unit. Why the morning freezes went unrecorded is considered under Risk Controls.

The freeze removed the primary means of position fixing named in the passage plan. While the plotter worked, the vessel had held the centreline. The heading began to change in the same second as the freeze, while the master was at the screen. His search for the beacon began 85 seconds later, and no position fix of any kind was taken in between.

It was extremely likely that the freeze led both to the unintended heading change and to the transit continuing without a fix.

No. 4 beacon screened by mangroves from the channel, 0.84 to 0.34 nm

The screening was measured, not modelled. The check was made from the port workboat on the inbound track, at the same time of night. The mangrove canopy on Merrandi Islet, about 240 m north of the beacon, stood 5.8 m high against a lantern height of 4.2 m, and the Authority's register had recorded the same screening since 09/2023. The eleven days between the occurrence and the check give no reason to doubt a mangrove canopy. The screening was virtually certain to have existed on the night.

With the plotter frozen, the beacon was the only reference left. The master said "Right. Where's four" at 18:08:31, when the vessel was 0.47 nm north of the beacon, and "There's four" at 18:10:02, when it was 0.25 nm north. In between, at about 0.40 nm, the vessel crossed the channel's western edge, inside the screened arc. Had the light been visible on the starboard bow, it would have shown the vessel drawing west within a minute of the turn, while the vessel was still in the channel.

Two points weigh against this. First, the heading change began before the search did, so the beacon's absence cannot explain the turn, only why it went unnoticed. Second, the light was clear of the islet from about 18:09:25, so it was visible for at least the last 37 seconds of the search. Because the vessel was west of the inbound track, the light would in fact have cleared the islet earlier still. Those seconds are discussed in the next subsection, but they do not alter the screening over most of the search.

It was likely that the screening of the beacon allowed the new heading to be held unnoticed until the vessel had left the channel. The deferral of the sightline clearing is considered under Organisational Influences.

Unshielded boat ramp floodlights aligned behind the No. 4 beacon from the north

The floodlights themselves are not in doubt. There were 14 unshielded LED floodlights on 12 m columns, each presenting about 4,800 candela toward the channel. From 0.5 nm north of the beacon, the array lay within 4 degrees of the beacon's bearing and within 1.5 degrees of its elevation. The array was continuously lit, whereas the beacon flashed green for 0.4 seconds every 2.5 seconds.

Some evidence suggested the floodlights delayed the sighting. The master described looking "into a wall of light" and called the beacon "a nothing". The outgoing master independently described it as "washed out altogether from up the top". For at least the last 37 seconds of the search, the light was clear of the islet and still not seen, so in that interval it was the floodlights, not the canopy, that stood between the master and the light.

Other evidence pointed the other way. For most of the search the beacon was screened and would not have been seen even against a dark background, and the turn began before the search. There was no measurement of what could be seen on the night, and nothing established a detection threshold for a 0.4 second green flash against that background. The master himself said twice that he could not tell whether the lights were why he could not see the beacon.

Settling the question would need a detection threshold for that flash against that background, determined from the vessel's own track. Whether the unshielded boat ramp floodlights aligned behind the No. 4 beacon delayed the sighting therefore remains unresolved, and nothing from it is carried into the findings.

Tide set down below prediction over Boolarong Bank

A tidal set-down was very unlikely. The port gauge read 0.91 m at 18:12 against a predicted 0.94 m, and at either height the vessel's effective draught of 1.53 m exceeded the depth of water over the bed at the contact point.

Master fatigued during the channel transit

Several points made fatigue plausible. The master was on day nine of a twelve-day tour, working a split duty of 05:40 to 08:40 and 13:00 to 19:20, with a spread of 13 hours 40 minutes. The transit fell in its last hour and a quarter. His diary showed 6 hours 15 minutes of sleep the night before, and he said in interview, "I was tired. I'd be lying if I said otherwise." After sighting the beacon, he eased the vessel back to port rather than stopping. He also believed he was just outside the channel edge when he was nearly 40 m west of the centreline, which is the kind of misjudgement fatigue is expected to produce.

The rest of the evidence, however, did not show impairment. In the same answer he said, "I wasn't fighting to stay awake and I wasn't slow," and his recorded performance agrees with that account. He held the centreline to within 3 m after entering the channel and responded to the shallow alarm in 4 seconds. His radio calls were complete and correct at the first attempt. He decided not to refloat in about thirty seconds, and that decision was correct. He also said that six and a quarter hours of sleep was normal for him on a split duty.

It was about as likely as not that the master was fatigued. A detailed fatigue assessment was not undertaken. Every decision that mattered was either explained by the absence of a night plan and hold point or contradicted by his recorded performance, so even an established fatigue would not have been shown to influence the occurrence. Whether the roster placed him in that position is considered under Organisational Influences. Whether the master was fatigued during the channel transit remains unresolved, and nothing from it is carried into the findings.

Individual Actions

Transit continued at eight knots from the plotter freeze until the shallow alarm

The vessel did not slow after the plotter froze at 18:07:06. Speed over the ground was 8.4 kt at the freeze and 8.1 kt at 18:08:00, and the later fall to 7.1 kt by 18:11:48 matched the drag of shoaling water. The engine recorder showed no throttle movement until 18:11:52. Since the recorded track, the engine data, the bridge audio and the master's account all agreed, it was virtually certain that the transit continued unchanged.

From the freeze, the master had no means of fixing the vessel's position. The bridge audio placed his attention first on the frozen screen and then, from 18:08:31, on the search for the No. 4 beacon. Nothing recorded showed that stopping was considered, and the master's account was the same. He went straight to the visual transit because, in his mind, it was the backup.

At 18:10:02 he saw the beacon dead ahead and knew he was west of the line. Rather than stopping, he brought the vessel back to port, because he believed he was just outside the edge. He was in fact 39 m west of the centreline, and the course over the ground came back only to 197 degrees, still west of the channel's 195. He said that stopping on the edge of a bank he could not see felt worse than easing back in.

Yet nothing in the conditions made stopping costly. There was no other traffic, the water was flat and the wind was offshore at 8 kt. The vessel's crash stop from 8 kt took 92 m. The vessel crossed the western edge 114 seconds after the freeze, and a stop begun in that time would have left it in the channel. Even later, a stop begun at any point before about 18:11:40 would have halted it short of the contact point, upright on the bank's margin and not holed. That window included the 98 seconds after the sighting.

No lookout was on the bridge, because the deckhand had been sent to collect fares at 18:04:14. This was normal practice on the route, and OPS-3 did not address it either way.

This action sat downstream of other factors. The passage plan had no hold point and no guidance for loss of the plotter, and the visual reference it relied on was screened for most of the search. Those matters are analysed under Risk Controls and Local Conditions.

It was extremely likely that continuing at about 8 kt without a position fix contributed to the vessel leaving the channel and reaching the bank.

Heading eased five degrees to starboard without a decision to turn

The heading began to change at 18:07:06, the second the plotter froze. It increased steadily from 195 to 200 degrees over about a minute and was then held. The rate of turn peaked at 6 degrees per minute at 18:07:40 and was back to zero by 18:08:50. Rudder angle followed helm demand within 1.5 degrees throughout, so the turn was steered. The master said in interview: "I eased her to starboard while I was fiddling with the plotter and then looking for four. Five degrees. I never made that decision." His safety report said the same.

The timing shows when the turn was made. Its peak came while the master was at the frozen screen, between the screen taps at 18:07:26 and "frozen again" at 18:07:44, and most of it was complete before the spoken search began at 18:08:31. It was therefore a turn that went unnoticed, not a turn made while looking for the light. Nothing on the bridge showed him the new heading, because the plotter's vessel symbol was stationary and the beacon was screened.

There was no current and no wind effect at that state of tide, so the westing that took the vessel out of the channel came from this heading change alone. Held at 200 degrees, the heading set the vessel west at about 20 m a minute and across the 20 m to the western edge by about 18:09:00. Whether the floodlights behind the beacon delayed the moment the heading was noticed could not be resolved, and that matter is discussed under Local Conditions.

It was virtually certain that the unintended five-degree heading change, made while the master's attention was on the frozen plotter, took the vessel out of the dredged channel.

Engines stopped and no attempt made to refloat on a falling tide

Both engines were stopped 27 seconds after contact and were not restarted. The pan pan at 18:13:04 stated "I am not attempting to refloat" as a decision already made. The master had the No. 2 starboard void sounded, reported about 75 mm of ingress at 18:17:05, and had the void sounded every ten minutes after that. Passengers were addressed 36 seconds after impact and again at 18:14:52, and all 38 were given lifejackets.

The survey later found a 180 mm split in the starboard plating in way of that void, and ingress reached approximately 260 mm by the time the vessel refloated at 00:20. The breach was in the compartment the vessel was resting on. Refloating into 3.2 m of water with the ingress unassessed would therefore have put a flooding compartment into deep water with passengers still aboard. The master gave four reasons for not refloating: the falling tide, the unknown bottom, the people aboard, and the fact that a vessel on the bank could not sink.

It was very likely that stopping the engines and not attempting to refloat reduced the risk to the passengers and the vessel after the grounding.

Plotter fault passed on verbally at the change of master and not entered in the defect book

The plotter froze and recovered itself at 10:42 and 11:17 that morning. At the change of master at about 13:05, the outgoing master told the relieving master about both freezes, and the relieving

master replied that it had done the same to him the month before. Neither of them entered the fault in the defect book or reported it to the Operations Manager. Both accounts agreed, and both masters volunteered the omission before being asked, so it was virtually certain that this occurred.

It was unlikely, however, that the omission influenced the occurrence. The freeze at 18:07 was a firmware fault, and an entry made at 13:05 would have been reviewed by the Operations Manager at the end of the week. Nothing written that afternoon could have applied the firmware fix or taken the vessel off TF18 by 18:07.

The omission was still a finding, because it reflected practice rather than a lapse. Both masters made the same omission and described the same rule for it, and OPS-9 did not require a fault that had cleared itself to be recorded. When a cleared fault on the primary means of position fixing is passed on by word of mouth, the next master knows about it but nobody ashore does. The procedure gap is analysed under Risk Controls.

It was likely that passing the cleared plotter fault on verbally, without a defect book entry, increased risk.

Occurrence Events

Vessel left the dredged channel on the bank side

The Merrandi Channel is 40 m wide and dredged to 3.2 m. The bank on its western side shoals at about 1 in 70 and dries in places. Once the vessel was west of the dredged edge, it was over the bank. The grounding followed directly from that departure, and from nothing else.

The recorded track fixed when and how the departure happened. The vessel held the centreline to within 3 m from the channel entrance until 18:07:06. From then, the heading increased from 195 to 200 degrees over about a minute and was held, and the vessel made about 20 m of westing a minute. It crossed the western edge at about 18:09:00, 0.40 nm north of the No. 4 beacon, and was over Boolarong Bank from about 18:11:00.

The shape of the movement showed what caused it. A current or a wind would have produced a drift with no turn behind it. Instead, the rate of turn peaked at 6 degrees per minute at 18:07:40 and was back to zero by 18:08:50, which is a heading change applied and then held. The wind was offshore at 8 kt, and there was no cross-set in the channel at that state of tide, so neither could account for movement of this kind. Why the heading changed, and why the transit continued at the same speed, are argued under individual actions.

At 18:10:02 the master said "There's four. Bring her back." The course over the ground then eased from 200 to 197 degrees. That slowed the westing but did not stop it, because the course stayed west of the channel's 195 degrees through to contact. By that point the vessel was already 39 m west of the centreline, 19 m beyond the western edge.

The surveyed grounding position, 74 m west of the charted centreline and 54 m outside the dredged channel, agreed with the track. So did the furrows in the bed, which ran on a heading of 197 degrees. It was virtually certain that the vessel left the dredged channel on its western side, and virtually certain that this departure led to the grounding.

Shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning

The echo sounder's shallow alarm was the vessel's last defence against the bank, and it worked exactly as it was set. The setting was 0.9 m under the keel, written into the setup at commissioning in 2019 and never changed. The alarm sounded once, at 18:11:48, and the alarm log, the bridge audio and the recorded speed agreed to the second. Both throttles went to astern 4 seconds later, and contact followed at 18:12:04.

The shortfall therefore lay in the setting and the speed, not in the equipment. In those 16 seconds the vessel covered 58 m at 7.1 kt, whereas the vessel's trials figure for a crash stop from 8 kt is 92 m. The alarm log noted that "at the surveyed bank gradient of about 1 in 70 across that section the alarm could not have sounded appreciably earlier at that setting." Astern thrust was still developing at the moment of contact. The warning was thus shorter than any stop the vessel could make, however quickly it was answered.

A different setting shows what margin was available. On the same 1 in 70 slope, a setting of about 1.6 m under the keel would have sounded about 107 m short of the contact, which allowed a crash stop with four seconds to react. Such a setting still would not have sounded in the dredged channel at low water, but a setting giving forty seconds' warning would have done so. The setting therefore has to be chosen for the channel and its gradients rather than simply raised.

The alarm sounded so late because the vessel was over the bank at transit speed, and that speed is argued under individual actions. It was virtually certain that the alarm, at its 2019 setting, gave 16 seconds' warning. It was extremely likely that this warning, being shorter than the vessel's stopping distance, contributed to the grounding.

Steering or propulsion failure before the vessel left the centreline

A steering or propulsion failure before the vessel left the centreline was very unlikely and was ruled out on two independent grounds:

- The engine recorder showed both engines within 12 rpm of each other, with no alarm or throttle movement until 18:11:52. Steering pressure held at 96 to 99 bar, rudder angle followed helm demand within 1.5 degrees, and the rate of turn showed a heading applied and held rather than a sheer.
- The steering system was removed and function tested on 12/07/2026 and was found serviceable, with all damage dating from the contact.

Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. It is virtually certain that section 4.2 of WAN-2 applied its availability, recording and notice to mariners duties only to aids on declared shipping channels. The Merrandi Channel was not one, so its users had no right to be told of a change to its beacons. This was not found to have contributed to the occurrence, and it is likely that it increased risk. (Safety issue)
2. Service TF18 was moved to 17:35 for the winter timetable from 01/06/2026, which put it in darkness. The change was assessed as a timetable change only, and RRA-07, which restricted the Merrandi Channel to daylight transits, was not reopened. It is likely that this contributed by leaving the channel with no night passage plan. Why the change was assessed that way was not established, and a reopened assessment might have produced a plan with no hold point in it. (Safety issue)
3. The Port Authority deferred work order WO-118, to clear the No. 4 beacon's sightline through the Merrandi Islet mangroves, as unfunded at the 2023, 2024 and 2025 works rounds, because the islet lay outside the 50 m vegetation scope of the services agreement. A trim funded in 2023 would have left the canopy near 3.5 m on 07/07/2026, below the 4.2 m lantern. It is very likely that the deferral contributed. (Safety issue)
4. In 11/2025 a solar conversion halved the No. 4 beacon's nominal range from 6 nm to 3 nm. No local notice to mariners was issued and no channel user was told. This was not found to have contributed, because the light was screened by mangroves where the vessel left the channel. It is likely that it increased risk. (Safety issue)
5. In the winter timetable the TF18 darkness transit closed a split duty with a 13 hour 40 minute spread, 20 minutes inside the operator's 14 hour limit, on every day of a twelve-day tour. This was not found to have contributed, because fatigue on the night was not established. It is likely that it increased risk. (Safety issue)
6. It is virtually certain that there was no night passage plan for the Merrandi Channel, and that the plan in use named no hold point or action for loss of the chart plotter, its primary means of position fixing. It is very likely that this absence contributed, because a stop begun before about 18:11:40 would have halted the vessel short of the contact point. (Safety issue)
7. OPS-9 did not require a fault that had cleared itself to be recorded. As a result, none of the chart plotter freezes the masters described in the year before the occurrence was written down, and the 11/2024 firmware addressing cause code 0x51 was never applied. It is likely that this contributed. How quickly a recorded freeze would have led to the fix was not established, given that the operator held no subscription to the manufacturer's bulletins. (Safety issue)
8. The forward saloon lifejacket locker could not be reached from 22 of the 34 forward saloon seats without passing behind a kiosk counter. The counter was fitted in 2019 under a minor modification notice that did not trigger a re-survey, and the operator reported the same counter on its two other 24 m vessels. This was not found to have contributed, and it is very likely that it increased risk. (Safety issue)
9. The chart plotter display froze at 18:07:06 during the channel transit and did not recover, which removed the primary means of position fixing. It is very likely that this contributed.

10. Between 0.84 nm and 0.34 nm north of the No. 4 beacon, the Merrandi Islet mangrove canopy screened the beacon from the channel. It is likely that the screening contributed by allowing the heading change to go unnoticed while the vessel left the channel. The screening did not cause the turn, which began at the frozen plotter. The light was also clear of the islet for at least the last 37 seconds before the master sighted it.

11. It is very unlikely that the tide set down below prediction over Boolarong Bank. The gauge read 0.91 m at 18:12 against a predicted 0.94 m, and the vessel would have touched at the predicted height as well.

12. The transit continued at about eight knots from the plotter freeze at 18:07:06 until the shallow alarm at 18:11:48, with no means of fixing the vessel's position. Nothing in the operator's procedures prompted a stop, although the water was flat, there was no other traffic and the vessel could have been stopped in 92 m. It is very likely that this contributed.

13. From 18:07:06 the heading increased from 195 to 200 degrees over about a minute and was then held. The master steered the turn without a decision to make it while he was at the frozen plotter, and nothing on the bridge showed him the new heading. This heading change contributed.

14. After contact, both engines were stopped within 27 seconds. Within a minute the master decided not to attempt to refloat the holed vessel on a falling tide with 38 passengers aboard. It is very likely that this reduced risk.

15. At the change of master at about 13:05, the outgoing master passed on the morning's two plotter freezes verbally, and neither master entered them in the defect book, in line with a practice both described. This was not found to have contributed, because nothing written that afternoon could have applied firmware to the unit by 18:07. It is likely that it increased risk.

16. The vessel crossed the western edge of the 40 m dredged channel at about 18:09 and was over Boolarong Bank from about 18:11, which resulted in the grounding.

17. The echo sounder shallow alarm remained at its 2019 commissioning setting of 0.9 m under the keel. It sounded 16 seconds before contact, in which time the vessel covered 58 m against a crash stop distance of 92 m from 8 kt. It is very likely that this contributed. The alarm itself functioned as set.

18. It is very unlikely that a steering or propulsion failure occurred before the vessel left the centreline. Engine speed, steering hydraulic pressure and rudder response were normal throughout, and every steering component was found serviceable.

Probable cause

The probable cause of the grounding was the continuation of the channel transit at about eight knots after the chart plotter display froze and did not recover, in the absence of any night passage plan or hold point for loss of the primary fix, which allowed a five degree heading change to starboard, made without a decision to turn, to take the vessel out of the dredged channel and onto Boolarong Bank.

Contributing factors

- It is virtually certain that the last service was moved into darkness without reopening the daylight-only risk assessment, and likely that this contributed. (Safety issue)
- It is virtually certain that there was no night passage plan or hold point for loss of the primary fix, and very likely that this absence contributed. (Safety issue)

- That the defect procedure never required a fault that cleared itself to be recorded is virtually certain, and it likely contributed. (Safety issue)
- The chart plotter display froze during the channel transit and did not recover, which is virtually certain, and this very likely contributed.
- With virtual certainty, the transit continued at eight knots from the plotter freeze until the shallow alarm, and this very likely contributed.
- Clearing of the No. 4 beacon's sightline was deferred unfunded for three works rounds, which is virtually certain, and the deferral very likely contributed. (Safety issue)
- It is virtually certain that the No. 4 beacon was screened by mangroves from the channel between 0.84 nm and 0.34 nm north of it, and likely that the screening contributed by leaving the heading change unnoticed.
- The heading eased five degrees to starboard without a decision to turn, and it is virtually certain both that this occurred and that it contributed.
- That the vessel left the dredged channel on the bank side is virtually certain, and so is its contribution.
- It is virtually certain that the shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning, and very likely that this contributed.

Other factors that increased risk

- The aids code set no notice or availability duty for beacons off declared channels, which is virtually certain. Although it was not found to have contributed to this occurrence, it likely increased risk. (Safety issue)
- It is virtually certain that the No. 4 beacon's range was halved by solar conversion with no notice to channel users. It was not found to have contributed to this occurrence, and it likely increased risk. (Safety issue)
- That the darkness transit was rostered to close a split duty within 20 minutes of the spread limit is virtually certain. It was not found to have contributed to this occurrence, and it is likely that it increased risk. (Safety issue)
- With virtual certainty, the forward saloon lifejacket locker was blocked by a 2019 counter fitted without re-survey. This was not found to have contributed to this occurrence, and it very likely increased risk. (Safety issue)
- The plotter fault was passed on verbally at the change of master and not entered in the defect book, which is virtually certain. This was not found to have contributed to this occurrence, but it likely increased risk.

Other findings

- It is very unlikely that the tide set down below prediction over Boolarong Bank, and this was ruled out.
- A steering or propulsion failure before the vessel left the centreline was very unlikely, and this was ruled out.
- It is virtually certain that the engines were stopped and no attempt was made to refloat on a falling tide, and very likely that this reduced risk.
- Fourteen unshielded floodlights at the new boat ramp, commissioned on 22/05/2026, lay behind the No. 4 beacon from the north, within 4 degrees of the same bearing from 0.5 nm. Whether they delayed the master's sighting of the beacon after it cleared the islet could not be determined either way. Both masters independently described the beacon as hard to pick out against the shore lighting. Against that, the canopy screened the

beacon over most of the search, and the heading change began before the search. The master himself said twice that he could not tell whether the lights were why he could not see the beacon. The matter could be settled only by a measure of what was visible on the night and of the detection threshold for a 0.4 second green flash against that background. Neither exists.

- Whether the master was fatigued during the channel transit could not be determined either way. He was on day nine of a twelve-day tour, had slept 6 hours 15 minutes and said he was tired. He also said he was not fighting to stay awake and was not slow. His recorded performance showed no sign of fatigue. He held the centreline within 3 m, responded to the shallow alarm in 4 seconds, completed his radio work correctly at the first attempt, and decided not to refloat in about thirty seconds. Even had fatigue been established, the decisions on which the occurrence turned are accounted for by the absence of a plan and hold point, or are contradicted by the recorded data.

Recommendations

Clearing of the beacon's sightline deferred unfunded for three works rounds

The port authority raised work order WO-118 in 09/2023 to trim mangrove regrowth on Merrandi Islet. The regrowth was screening the No. 4 beacon from the northern approach. The work was deferred at the 2023, 2024 and 2025 works rounds, and each entry reads "defer, unfunded, refer next round". The islet lies 240 m from the beacon, outside the 50 m vegetation scope of schedule 2.4 of the aids to navigation services agreement. The work therefore competed for the general waterways vote rather than the aids to navigation vote.

On the night of the grounding the canopy stood 1.6 m above the 4.2 m lantern. A trim funded at the 2023 round would very likely have left it about 0.7 m below the lantern. The beacon would then have been open to the channel over the stretch where it was screened.

Last service moved into darkness without reopening the daylight-only risk assessment

The operator moved its last Merrandi Channel service into darkness without reopening the route risk assessment that restricted the channel to daylight. RRA-07 issue 2 permits a night transit only with the General Manager Operations' approval and a specific passage plan. CHG-2026-014 moved TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026. The Operations Manager approved it as a timetable change only. Neither the approval nor the plan that RRA-07 requires was obtained.

The issue matters because the change process tested whether a change was new. It did not test whether the existing assessment still held. A restriction in a controlled risk assessment could therefore be set aside by a change that never engaged with it. As a result, a 96-passenger vessel crossed a 40 m channel in darkness 31 times without a night passage plan. Approving the change this way likely contributed to there being no night passage plan and no hold point for loss of the primary fix. A reopened assessment might have produced a plan with no such hold point, but that bears on the quality of the plan, not on whether one would have existed. What the Operations Manager understood the restriction to require of him was not established, so the reason the change was assessed as a timetable change only is not known.

No safety action has been recorded against this safety issue.

No night passage plan or hold point for loss of the primary fix

The Merrandi Channel passage plan was written for the daylight transit and stated that night transits were not scheduled. The operator had no night passage plan for the channel. The plan named the chart plotter as the primary means of position fixing but said nothing about what a master should do if it was lost during the transit. It named no abort criterion or hold point.

This was the operator's own defence against the sequence that occurred, and it had not been written. The absence of the plan and of a hold point very likely contributed to the transit continuing at eight knots after the plotter froze. At the point where the plotter was lost, the No. 4 beacon was the whole of the secondary means. The town wharf leading lights did not resolve as two lights until the vessel was inside the channel's southern third. A hold point costs nothing on a flat night with no traffic. Both masters interviewed said independently that stopping was the right response and that nothing in the operator's procedures asked for it.

Defect procedure never requires a fault that cleared itself to be recorded

The operator's defect procedure in Operations Manual OPS-9 contained no requirement to record a fault that had cleared itself. Section 9.1 required immediate reporting only of a defect that rendered the vessel unseaworthy or affected a statutory item. The chart plotter was neither, although the passage plan named it as the primary means of position fixing. Section 9.2 left all other defects to the defect book at the end of the shift. For a fault that had already come and gone, that requirement never applied. Section 9.3 required a verbal brief at the change of master. It did not require the brief to be written down, and it required nothing of the relieving master afterwards. Both masters independently described the practice that followed: "If it's broken you write it, if it came good you say it." It is virtually certain that this gap existed.

The gap mattered because it kept an intermittent fault on the primary means of position fixing from ever reaching anyone ashore. On the outgoing master's account, the plotter froze several times in the year before the occurrence and twice on the morning of 07/07/2026. None of those freezes was written down. A single bench test by the contractor later identified the cause. It is likely that the gap contributed to the display freezing at 18:07 and not recovering. It could not be established how many recorded freezes would have been needed before the unit was sent for repair, or how quickly that would have happened. This explains the omission at the change of master. It does not excuse it.

No safety action has been recorded against this issue.

Aids code set no notice or availability duty for beacons off declared channels

The Waterways Aids to Navigation Code WAN-2 places its availability, recording and notification duties only on aids that stand on a declared shipping channel. The Merrandi Channel is not one. Its beacons fall under section 4.3, which requires only that the responsible authority can demonstrate an inspection interval it has set itself. Neither section 4.3 nor the services agreement that sets that interval requires availability to be recorded. Neither requires any user of the channel to be told of a defect, a deferral or a change.

This gap did not influence the grounding. It was nonetheless likely to be important to safety on the channel. The channel carries a 96-passenger ferry in darkness. The code gives its operator no right to be told when a beacon's range, character or position changes, and it provides no figure for how often the channel's aids work.

The solar conversion of the No. 4 beacon in 11/2025 halved its nominal range. Because of this gap, the change reached nobody who used the channel for eight months. A future change to a light that is not screened would reach nobody either. The Port Authority can close this gap through its own practice, whatever the code requires.

No safety action has been recorded against this issue.

Beacon's range halved by solar conversion with no notice to channel users

In 11/2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The new lantern is rated at a nominal 3 nautical miles, against 6 for the lantern it replaced. Its character, Fl G 2.5s, did not change. A measurement from the port workboat on 18/07/2026 gave a luminous range of 2.6 nm. No local notice to mariners

was issued and no operator using the channel was told. A notification lodged with the hydrographic service on 04/02/2026 had not been reflected in the chart by 07/07/2026, and the operator's passage plan appendix still gave the beacon at 6 M.

It is unlikely that this contributed to the grounding. Over the stretch where the vessel left the channel the beacon was screened by mangroves, and a brighter light behind the canopy would still have been screened. It is likely that it increased risk. An aid on a 96-passenger route lost half its nominal range, and the change reached nobody who relied on it. On a night when the light is not screened, or where another aid is changed the same way, a master working to the plan would be looking for a light twice as strong as the one there.

Local notice LNM-2026-14 has been issued and the Port Authority's aids to navigation register has been released. This action is in progress. It addresses the issue in part. [NOT ESTABLISHED: whether the chart correction has since been completed, and whether future changes to aids on the channel will be notified to its users.] No recommendation is made beyond this action.

Darkness transit rostered to close a split duty within 20 minutes of the spread limit

The winter timetable places the TF18 darkness transit at the end of a split duty. The duty runs from 05:40 to 08:40 and from 13:00 to 19:20, which gives 9 hours 20 minutes of duty across a 13 hour 40 minute spread. The operator's limits are 10 hours of duty and 14 hours of spread. The transit therefore falls in the last hour and a quarter of a spread that ends 20 minutes inside the limit. The master on that line works this pattern on every day of a 12 days on, 2 days off tour. No relief master is rostered. A master who reports unfit is covered by extending another master's tour.

It was unlikely that the roster influenced this occurrence. The roster could have reached the occurrence only through fatigue, and the master's recorded performance on the night showed no sign of it.

The issue remains a safety issue because it does not depend on this one night. It is a standing feature of the winter roster, and it recurs on every darkness transit in the timetable. On days nine to twelve of each tour, the last darkness transit of the day is still made at the end of a near-limit spread. The fatigue diary the operator asks for does not change this, because it is not acted on. There is also no relief master to break a tour.

No safety action has been recorded against this safety issue.

Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey

On MV Tarrakin Star, most of the forward saloon could not reach the lifejacket locker that served it. A kiosk counter fitted in a 2019 interior refit stood in front of the locker. The refit was carried out under a minor modification notice, and that notice did not trigger a re-survey of the safety equipment arrangement. The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear.

The survey of 16/07/2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter. With people standing at the counter, a 0.6 m inboard gap was the only route. The muster placard directed passengers to a locker they could not see from their seats. While issuing lifejackets, the deckhand had to ask three passengers to move away from the counter before passing behind it.

The obstruction played no part in the grounding or its outcome. It very likely increased risk nonetheless. On this occasion the vessel was aground, upright and stable, and the muster was unhurried. Any one of the following would have made the obstruction critical:

- a fire
- a capsize risk
- a night evacuation into the water
- a deckhand injured in the impact

In any of these, passengers who could not reach the lifejackets without crew help would have been left waiting. The counter remained in place on the vessel. The operator stated that the same counter is fitted to its two other 24 m vessels.

No safety action has been recorded against this issue.

Distribution

Title	Name
General Manager Operations	H. Vardon
Safety Manager	P. Nandasiri
Operations Manager	C. Bickerton
Harbour Master, Kanooka Heads	L. Truscott
Aids to Navigation Officer, Kanooka Shire Council	E. Maguire